Project Presentation Plan
An Era of the Seas

I. Presentation of the project (1min - Joseph)

II. Planning (2min)
A) Common Tasks (1min - Ethan)
B) Tasks of each member (1min - Achref)

III. Individual Tasks (8min)
A) Jérôme (2min)
Seach & methods
Conception
B) Achref (2min)
Seach & methods
Conception
C) Ethan (2min)
Seach & methods
Conception
D) Joseph (2min)
Seach & methods
Conception

IV. Conclusion (1min – Jérôme)
Tasks done
Tasks in progress
Tasks to do

I. Presentation of the project

The *An Era of the Seas* project stands out for its ambition to create a captivating maritime world where adventure and exploration are at the heart of the gaming experience. Developed by the passionate team at Dream Chasers, this visual novel invites player to immerse themselves in the fascinating universe of sailing through epic quests and strategic challenges.

II. Planning

A) Common Tasks

For this project all members have crucial common tasks.

First, the most important task is the testing, because we must be sure that our code and implementations work as expected.

Second, all members should make the most optimized code possible to make the most efficient execution of the game.

Both tasks are important to have the least bug possible to avoid compromising the gaming experience of the player.

B) Distribution of tasks

Jérôme oversees the Menu and UI, the implementation of the items, the network, and the music.
Achref is in charge of the simulation of the weather AI, the fishing system, the website and the design of entities.
Ethan is in charge of the simulation of water, the map, the system of hunger, the implementation of the quests and the animation of the entities.
Finally, Joseph is in charge of the movement of the player and the boat, the location system, the multiplayer, and the story.

III. Individual Tasks

A) Jérôme

Seach & methods

To implement the items, I listed all the items of the game, then I defines the statistic of each item into a csv file.

I also wrote the items information in a Json file such as the name, rarity, maxlevel.

Now we have to load those data in a script in Unity. Which is why I used Newtonsoft.Json and CsvHelper. To be able to use it, I searched on the documentation and on YouTube.

For the menu and UI, I mostly watched on YouTube how to manipulate UI elements on Unity.

Conception

B) Achref

Seach & methods

To begin, I tried to design the Dream Chasers website from scratch based on my knowledge using some basic HTML/CSS.

While creating this website, I wondered how it will get deployed, that is when I thought of using Odoo to get it deployed. So, I tried to paste my code on the application, but it asked me to redesign the website from scratch using their editorial tools.

As for the AI part, I started by writing down how my AI should behave, it's Patrol Mode and it's Chase Mode. Then with the help of some YouTube tutorials on AI in unity, I ended up by creating an AI that would be "casually" walking and whenever the player gets too close to it. It will start chasing until he runs away or gets out of sight.

Conception

C) Ethan

The first thing that I focused on is the water simulation.
Initially, the water simulation used a simple sine wave model. The WaveManager script calculated wave heights based on a simple sine function, which was then applied to the vertices of the water mesh through the WaterManager script. This method allowed us to create basic waves, while the Floater script handled buoyancy by adjusting the forces on floating objects according to the wave height. However, this model felt unrealistic and way too mechanical. The waves felt too uniform and static, especially for larger scenes like oceans or lakes. So instead, after reading an article on https://catlikecoding.com/ I adopeted a new model: Gerstner waves.

Gerstner waves work by calculating the horizontal and vertical displacement of water vertices. Instead of just moving vertices up and down like sine waves, Gerstner waves push them along a curved path, creating the illusion of rolling water. This is achieved using a mathematical function that combines wave steepness, direction, wavelength, and speed.

This model made the simulation way more lively.

D) Joseph

Seach & methods

For the movement,

- Classic Character Controller in Unity, helped player by tutorials on YouTube.

For the boat movement,

- I designed a helm and an interaction system (thanks to Ray Casting)
- Thanks to what I learned with Player Movement, I used the same principles in Boat Movement, but checking if the player was behind the Helm, and adjusting the speed to a more realistic one.
- Anchor Boolean to stop the boat

For Wind Simulation,

- I looked at real data about the best angle to yield the most force from a sail and replicated it in game.

Conception

IV. Conclusion

Overall, the Dream Chasers has managed to finish the Menu and UI, Implementation of items, the website, the 2D map, the water simulation and the movement of the boat and the player.

The team is currently working on the AI for the movement, the 3D map, the network.

We still have to do the weather simulation, the system of localization, fishing and hunger, the multiplayer, the story quest, the design and animation of the entities and the music of the game.